E5 Trigonometric identities

Name: ______________________

Class: ______________________

Date: ______________________

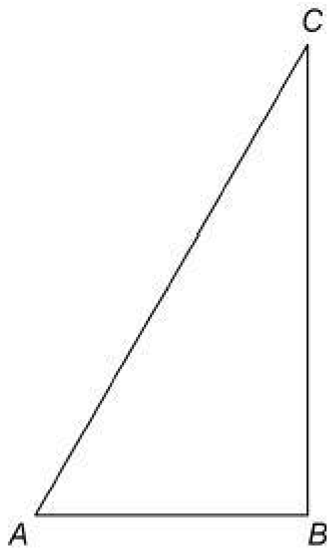
Time: **359 minutes**

Marks: **297 marks**

Comments:

Q1.

ABC is a right-angled triangle.



D is the point on hypotenuse *AC* such that *AD* = *AB*.

The area of ΔABD is equal to half that of ΔABC.

(a) Show that $\tan A = 2 \sin A$

(4)

(b) (i) Show that the equation given in part (a) has two solutions for $0° \leq A \leq 90°$

(2)

(ii) State the solution which is appropriate in this context.

(1)

(Total 7 marks)

Q2.

It is given that $\cos 15° = \frac{1}{2}\sqrt{2+\sqrt{3}}$ and $\sin 15° = \frac{1}{2}\sqrt{2-\sqrt{3}}$

Show that $\tan^2 15°$ can be written in the form $a + b\sqrt{3}$, where a and b are integers.

Fully justify your answer.

(Total 3 marks)

Q3.

$p(x) = 30x^3 - 7x^2 - 7x + 2$

(a) Prove that $(2x + 1)$ is a factor of $p(x)$

(2)

(b) Factorise $\mathrm{p}(x)$ completely.

(3)

(c) Prove that there are no real solutions to the equation

$$\frac{30\sec^2 x + 2\cos x}{7} = \sec x + 1$$

(5)
(Total 10 marks)

Q4.

Solve the equation

$$\sin\theta\tan\theta + 2\sin\theta = 3\cos\theta \quad \text{where } \cos\theta \neq 0$$

Give **all** values of θ to the nearest degree in the interval $0° < \theta < 180°$
Fully justify your answer.

(Total 5 marks)

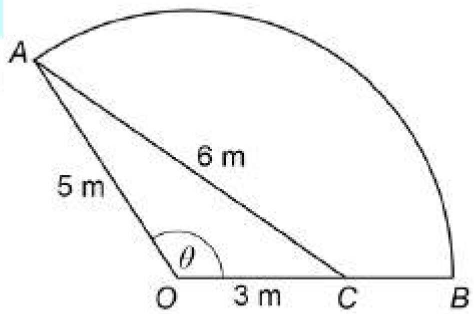
Q5.

Prove the identity $\cot^2\theta - \cos^2\theta \equiv \cot^2\theta\cos^2\theta$

(Total 3 marks)

Q6.

A wooden frame is to be made to support some garden decking. The frame is to be in the shape of a sector of a circle. The sector *OAB* is shown in the diagram, with a wooden plank *AC* added to the frame for strength. *OA* makes an angle of θ with *OB*.



(a) Show that the exact value of $\sin\theta$ is $\frac{4\sqrt{14}}{15}$

(3)

(b) Write down the value of θ in radians to 3 significant figures.

(1)

(c) Find the area of the garden that will be covered by the decking.

(2)

(Total 6 marks)

Q7.

(a) Write down the two solutions of the equation $\tan(x + 30°) = \tan 79°$ in the interval $0° \leq x \leq 360°$.

(2)

(b) Describe a single geometrical transformation that maps the graph of $y = \tan x$ onto the graph of $y = \tan(x + 30°)$.

(2)

(c) (i) Given that $5 + \sin^2 \theta = (5 + 3 \cos \theta) \cos \theta$, show that $\cos \theta = \frac{3}{4}$.

(5)

(ii) **Hence** solve the equation $5 + \sin^2 2x = (5 + 3 \cos 2x) \cos 2x$ in the interval $0 < x < 2\pi$, giving your values of x in radians to three significant figures.

(3)

(Total 12 marks)

Q8.

(a) Show that

$$\frac{\sec^2 x}{(\sec x + 1)(\sec x - 1)}$$

can be written as $\operatorname{cosec}^2 x$.

(3)

(b) Hence solve the equation

$$\frac{\sec^2 x}{(\sec x + 1)(\sec x - 1)} = \operatorname{cosec} x + 3$$

giving the values of x to the nearest degree in the interval $-180° < x < 180°$.

(6)

(c) Hence solve the equation

$$\frac{\sec^2(2\theta - 60°)}{(\sec(2\theta - 60°) + 1)(\sec(2\theta - 60°) - 1)} = \operatorname{cosec}(2\theta - 60°) + 3$$

giving the values of θ to the nearest degree in the interval $0° < \theta < 90°$.

(2)

(Total 11 marks)

Q9.

(a) Express $3 \cos x + 2 \sin x$ in the form $R \cos(x - \alpha)$, where $R > 0$ and $0° < \alpha < 90°$, giving your value of α to the nearest 0.1°.

(3)

(ii) Hence find the minimum value of $3 \cos x + 2 \sin x$ and the value of x in the interval
$0° < x < 360°$ where the minimum occurs. Give your value of x to the nearest 0.1°.

(3)

(b) (i) Show that $\cot x - \sin 2x = \cot x \cos 2x$ for $0° < x < 180°$.

(3)

(ii) Hence, or otherwise, solve the equation

$$\cot x - \sin 2x = 0$$

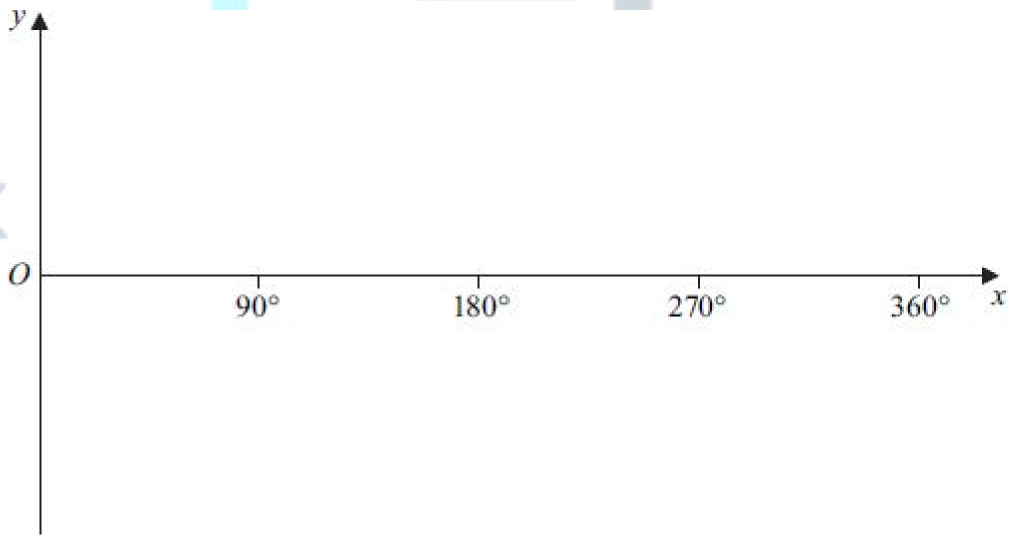
in the interval $0° < x < 180°$.

(3)

(Total 12 marks)

Q10.

(a) (i) On the axes given below, sketch the graph of $y = \tan x$ for $0° \leq x \leq 360°$.



(3)

(ii) Solve the equation $\tan x = -1$, giving all values of x in the interval $0° \leq x \leq 360°$.

(2)

(b) (i) Given that $6 \tan \theta \sin \theta = 5$, show that $6 \cos^2 \theta + 5 \cos \theta - 6 = 0$.

(3)

(ii) **Hence** solve the equation $6 \tan 3x \sin 3x = 5$, giving all values of x to the nearest degree in the interval $0° \leq x \leq 180°$.

(6)
(Total 14 marks)

Q11.

By forming and solving a quadratic equation, solve the equation

$$8 \sec x - 2 \sec^2 x = \tan^2 x - 2$$

in the interval $0 < x < 2\pi$, giving the values of x in radians to three significant figures.

(Total 7 marks)

Q12.

(a) Given that $2 \sin \theta = 7 \cos \theta$, find the value of $\tan \theta$.

(2)

(b) (i) Use an appropriate identity to show that the equation

$$6 \sin^2 x = 4 + \cos x$$

can be written as

$$6 \cos^2 x + \cos x - 2 = 0$$

(2)

(ii) Hence solve the equation $6 \sin^2 x = 4 + \cos x$ in the interval $0° < x < 360°$, giving your answers to the nearest degree.

(6)
(Total 10 marks)

Q13.

(a) By using a suitable trigonometrical identity, solve the equation

$$\tan^2 \theta = 3(3 - \sec \theta)$$

giving all solutions to the nearest 0.1° in the interval $0° < \theta < 360°$.

(6)

(b) Hence solve the equation

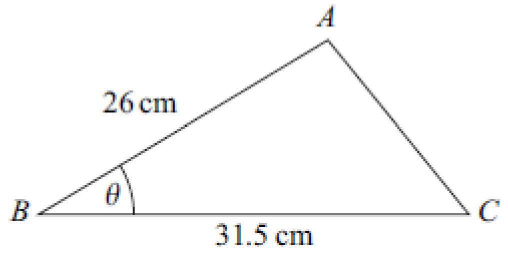
$$\tan^2(4x - 10°) = 3[3 - \sec(4x - 10°)]$$

giving all solutions to the nearest 0.1° in the interval $0° < x < 90°$.

(3)
(Total 9 marks)

Q14.

The triangle ABC, shown in the diagram, is such that AB = 26 cm and BC = 31.5 cm.



The acute angle ABC is θ, where $\sin\theta = \frac{5}{13}$.

(a) Calculate the area of triangle ABC.

(2)

(b) Find the exact value of $\cos\theta$.

(1)

(c) Calculate the length of AC.

(3)

(Total 6 marks)

Q15.

It is given that $(\tan\theta + 1)(\sin^2\theta - 3\cos^2\theta) = 0$.

Find the possible values of $\tan\theta$.

(Total 4 marks)

Q16.

(a) Show that the equation

$$\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} = 32$$

can be written in the form

$$\operatorname{cosec}^2\theta = 16$$

(4)

(b) Hence, or otherwise, solve the equation

$$\frac{1}{1+\cos(2x-0.6)} + \frac{1}{1-\cos(2x-0.6)} = 32$$

giving all values of x in radians to two decimal places in the interval $0 < x < \pi$.

(5)

(Total 9 marks)

Q17.

(a) Given that $x = \dfrac{\sin y}{\cos y}$, use the quotient rule to show that

$$\frac{dx}{dy} = \sec^2 y$$

(3)

(b) Given that $\tan y = x - 1$, use a trigonometrical identity to show that

$$\sec^2 y = x^2 - 2x + 2$$

(2)

(c) Show that, if $y = \tan^{-1}(x - 1)$, then

$$\frac{dy}{dx} = \frac{1}{x^2 - 2x + 2}$$

(1)

(d) A curve has equation $y = \tan^{-1}(x - 1) - \ln x$.

(i) Find the value of the x-coordinate of each of the stationary points of the curve.

(4)

(ii) Find $\dfrac{d^2y}{dx^2}$.

(2)

(iii) Hence show that the curve has a minimum point which lies on the x-axis.

(2)

(Total 14 marks)

Q18.

(a) Solve the equation $\tan x = -3$ in the interval $0° \le x \le 360°$ giving your answers to the nearest degree.

(3)

(b) (i) Given that

$$7 \sin^2 \theta + \sin \theta \cos \theta = 6$$

show that

$$\tan^2 \theta + \tan \theta - 6 = 0$$

(3)

(ii) Hence solve the equation $7\sin^2\theta + \sin\theta\cos\theta = 6$ in the interval $0° \le \theta \le 360°$, giving your answers to the nearest degree.

(4)
(Total 10 marks)

Q19.

(a) Solve the equation $\sec x = -5$, giving all values of x in radians to two decimal places in the interval $0 < x < 2\pi$.

(3)

(b) Show that the equation

$$\frac{\operatorname{cosec} x}{1+\operatorname{cosec} x} - \frac{\operatorname{cosec} x}{1-\operatorname{cosec} x} = 50$$

can be written in the form

$$\sec^2 x = 25$$

(4)

(c) Hence, or otherwise, solve the equation

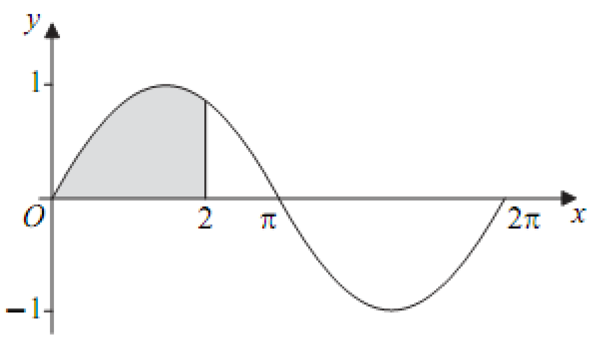
$$\frac{\operatorname{cosec} x}{1+\operatorname{cosec} x} - \frac{\operatorname{cosec} x}{1-\operatorname{cosec} x} = 50$$

giving all values of x in radians to two decimal places in the interval $0 < x < 2\pi$.

(3)
(Total 10 marks)

Q20.

A curve C, defined for $0 \le x \le 2\pi$ by the equation $y = \sin x$, where x is in radians, is sketched below. The region bounded by the curve C, the x-axis from 0 to 2 and the line $x = 2$ is shaded.



(a) The area of the shaded region is given by $\int_0^2 \sin x \, dx$, where x is in radians.

Use the trapezium rule with five ordinates (four strips) to find an approximate value for the area of the shaded region, giving your answer to three significant figures.

(4)

(b) Describe the geometrical transformation that maps the graph of $y = \sin x$ onto the graph of $y = 2 \sin x$.

(2)

(c) Use a trigonometrical identity to solve the equation

$$2 \sin x = \cos x$$

in the interval $0 \le x \le 2\pi$, giving your solutions in radians to three significant figures.

(4)

(Total 10 marks)

Q21.

Prove that, for all values of x, the value of the expression

$$(3 \sin x + \cos x)^2 + (\sin x - 3 \cos x)^2$$

is an integer and state its value.

(Total 4 marks)

Q22.

(a) (i) Solve the equation $\operatorname{cosec}\theta = -4$ for $0° < \theta < 360°$, giving your answers to the nearest 0.1°.

(2)

(ii) Solve the equation

$$2 \cot^2(2x + 30°) = 2 - 7 \operatorname{cosec}(2x + 30°)$$

for $0° < x < 180°$, giving your answers to the nearest 0.1°.

(6)

(b) Describe a sequence of two geometrical transformations that maps the graph of $y = \operatorname{cosec} x$ onto the graph of $y = \operatorname{cosec}(2x + 30°)$.

(4)

(Total 12 marks)

Q23.

(a) Solve the equation $\tan(x + 52°) = \tan 22°$, giving the values of x in the interval $0° \le x \le 360°$.

(3)

(b) (i) Show that the equation

$$3\tan\theta = \frac{8}{\sin\theta}$$

can be written as

$$3\cos^2\theta + 8\cos\theta - 3 = 0$$

(3)

(ii) Find the value of $\cos\theta$ that satisfies the equation

$$3\cos^2\theta + 8\cos\theta - 3 = 0$$

(2)

(iii) Hence solve the equation

$$3\tan 2x = \frac{8}{\sin 2x}$$

giving all values of x to the nearest degree in the interval $0° \le x \le 180°$.

(4)
(Total 12 marks)

Q24.

(a) Solve the equation

$$\operatorname{cosec} x = 3$$

giving all values of x in radians to two decimal places, in the interval $0 \le x \le 2\pi$.

(2)

(b) By using a suitable trigonometric identity, solve the equation

$$\cot^2 x = 11 - \operatorname{cosec} x$$

giving all values of x in radians to two decimal places, in the interval $0 \le x \le 2\pi$.

(6)
(Total 8 marks)

Q25.

It is given that $y = \tan 4x$.

(a) By writing $\tan 4x$ as $\frac{\sin 4x}{\cos 4x}$, use the quotient rule to show that

$\frac{dy}{dx} = p(1 + \tan^2 4x)$, where p is a number to be determined.

(3)

(b) Show that $\frac{d^2y}{dx^2} = qy(1 + y^2)$, where q is a number to be determined.

(5)

(Total 8 marks)

Q26.

(a) Sketch the graph of $y = \cos x$ in the interval $0 \leq x \leq 2\pi$. State the values of the intercepts with the coordinate axes.

(2)

(b) (i) Given that

$$\sin^2 \theta = \cos \theta(2 - \cos \theta)$$

prove that $\cos \theta = \frac{1}{2}$.

(2)

(ii) Hence solve the equation

$$\sin^2 2x = \cos 2x(2 - \cos 2x)$$

in the interval $0 \leq x \leq \pi$, giving your answers in radians to three significant figures.

(4)

(Total 8 marks)

Q27.

(a) Show that the equation

$$10 \operatorname{cosec}^2 x = 16 - 11 \cot x$$

can be written in the form

$$10 \cot^2 x + 11 \cot x - 6 = 0$$

(1)

(b) Hence, given that $10 \operatorname{cosec}^2 x = 16 - 11 \cot x$, find the possible values of $\tan x$.

(4)

(Total 5 marks)

Q28.

(a) Solve the equation $\sec x = \frac{3}{2}$, giving all values of x to the nearest degree in the interval $0° < x < 360°$.

(2)

(b) By using a suitable trigonometrical identity, solve the equation

$$2\tan^2 x = 10 - 5\sec x$$

giving all values of x to the nearest degree in the interval $0° < x < 360°$.

(6)
(Total 8 marks)

Q29.

(a) Given that $\dfrac{\sin\theta - \cos\theta}{\cos\theta} = 4$, prove that $\tan\theta = 5$.

(2)

(b) (i) Use an appropriate identity to show that the equation

$$2\cos^2 x - \sin x = 1$$

can be written as

$$2\sin^2 x + \sin x - 1 = 0$$

(2)

(ii) Hence solve the equation

$$2\cos^2 x - \sin x = 1$$

giving all solutions in the interval $0° \le x \le 360°$.

(5)
(Total 9 marks)

Q30.

(a) Solve the equation $\tan x = -\dfrac{1}{3}$, giving all the values of x in the interval $0 < x < 2\pi$ in radians to two decimal places.

(3)

(b) Show that the equation

$$3\sec^2 x = 5(\tan x + 1)$$

can be written in the form $3\tan^2 x - 5\tan x - 2 = 0$.

(1)

(c) Hence, or otherwise, solve the equation

$$3\sec^2 x = 5(\tan x + 1)$$

giving all the values of x in the interval $0 < x < 2\pi$ in radians to two decimal places.

(4)

(Total 8 marks)

Q31.

(a) Given that

$$\frac{3+\sin^2\theta}{\cos\theta-2}=3\cos\theta$$

show that

$$\cos\theta=-\frac{1}{2}$$

(4)

(b) Hence solve the equation

$$\frac{3+\sin^2 3x}{\cos 3x-2}=3\cos 3x$$

giving all solutions in degrees in the interval $0° < x < 180°$.

(4)

(Total 8 marks)

Q32.

(a) Solve the equation $\cot x = 2$, giving all values of x in the interval $0 \leq x \leq 2\pi$ in radians to two decimal places.

(2)

(b) Show that the equation $\operatorname{cosec}^2 x = \dfrac{3\cot x+4}{2}$ can be written as

$$2\cot^2 x - 3\cot x - 2 = 0$$

(2)

(c) Solve the equation $\operatorname{cosec}^2 x = \dfrac{3\cot x+4}{2}$, giving all values of x in the interval $0 \leq x \leq 2\pi$ in radians to two decimal places.

(4)

(Total 8 marks)

Q33.

(a) Solve the equation $\sin 2x = \sin 48°$, giving the values of x in the interval $0° \leq x < 360°$.

(4)

(b) Solve the equation $2\sin\theta - 3\cos\theta = 0$ in the interval $0° \leq \theta < 360°$, giving your answers to the nearest 0.1°.

(4)

(Total 8 marks)

Q34.

(a) Solve the equation sec x = 3, giving the values of x in radians to two decimal places in the interval $0 \leq x < 2\pi$.

(3)

(b) Show that the equation $\tan^2 x = 2 \sec x + 2$ can be written as $\sec^2 x - 2 \sec x - 3 = 0$.

(2)

(c) Solve the equation $\tan^2 x = 2 \sec x + 2$, giving the values of x in radians to two decimal places in the interval $0 \leq x < 2\pi$.

(4)

(Total 9 marks)

Q35.

(a) Given that $y = \frac{\sin\theta}{\cos\theta}$, use the quotient rule to show that $\frac{dy}{d\theta} = \sec^2 \theta$.

(3)

(b) Given that $x = \sin \theta$, show that $\frac{x}{\sqrt{1-x^2}} = \tan\theta$.

(2)

(c) Use the substitution $x = \sin \theta$ to find $\int \frac{1}{(1-x^2)^{\frac{3}{2}}}\,dx$, giving your answer in terms of x.

(5)

(Total 10 marks)